

Respiratory infections

Upper Respiratory Tract Infections

- involve the nasal cavity and pharynx
- named according to the anatomic sites primarily involved
- **Rhinitis (or coryza)** implies inflammation of the nasal mucosa
- **Pharyngitis** denotes pharyngeal infection
- **Tonsillitis** indicates an inflammatory involvement of the tonsils
- infections may simultaneously involve two or more sites (eg, rhinopharyngitis or tonsillopharyngitis)

Cont...

Clinical features

Rhinitis

- most common manifestation of the common cold
- variable fever, inflammatory edema of the nasal mucosa, and an increase in mucous secretions
- varying degrees of nasal obstruction
- the nasal discharge may be clear and watery at the onset of illness, becoming thick and sometimes purulent as the infection progresses over 5 to 10 days

Cont...

Pharyngitis and tonsillitis

- pharyngeal pain (sore throat) and the clinical appearance of erythema and swelling of the affected tissues

COMMON ETIOLOGIC AGENTS

- most (more than 80%) are caused by viruses
- most frequent bacterial cause
 - *S. pyogenes*
- *Corynebacterium diphtheria*
 - a major pathogen that continues to cause infection in many countries
 - clinical and epidemiologic findings suggest

Cont...

- *Neisseria gonorrhoeae*
 - a pharyngeal pathogen that is usually transmitted by oral–genital contact
- other bacteria have been implicated as causes of acute pharyngitis (eg, *Corynebacterium ulcerans*, *Arcanobacterium haemolyticum*, *Francisella tularensis*, and streptococci of groups B, C, and G)

Major Infectious Causes of Upper Respiratory Disease

DISEASE	VIRUSES	BACTERIA AND FUNGI
Rhinitis	Rhinoviruses, adenoviruses, coronaviruses, parainfluenza viruses, influenza viruses, respiratory syncytial virus, some coxsackie A viruses	Rare
Pharyngitis or tonsillitis	Adenoviruses, parainfluenza viruses, influenza viruses, rhinoviruses, coxsackie A or B virus, herpes simplex virus, Epstein–Barr virus	Group A streptococcus (<i>S. pyogenes</i>), <i>Corynebacterium diphtheriae</i> , <i>Neisseria gonorrhoeae</i>

Rhinoviruses

- Picornaviridae
- Rhinovirus
- Virion 28-30 nm - non enveloped
- Icosahedral symmetry
- Four capsid proteins (VP1-VP4)
- One or two nonstructural proteins
- Genome : ss RNA, linear, positive sense
- Replication and assembly in cytoplasm
- Insensitive to ether, chloroform and nonionic detergents
- acid labile

Cont...

- Common cold viruses
- Mild URT illnesses
- Nasal secretions, throat and oral secretions
- More thermo stable than enteroviruses and survive for days on environmental surfaces
- Infect only humans, chimpanzee
- Replication is limited to the epithelium of the nasal mucosa
- Human cell lines - grow better at 33⁰c (the temperature of the nasopharynx in human)
- More than 100 serotypes

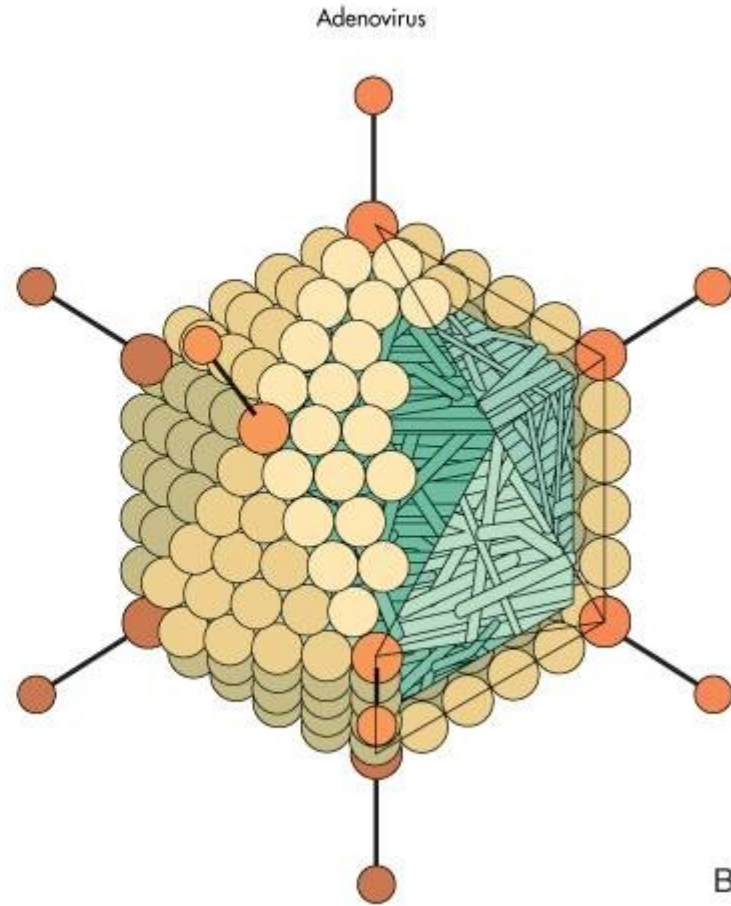
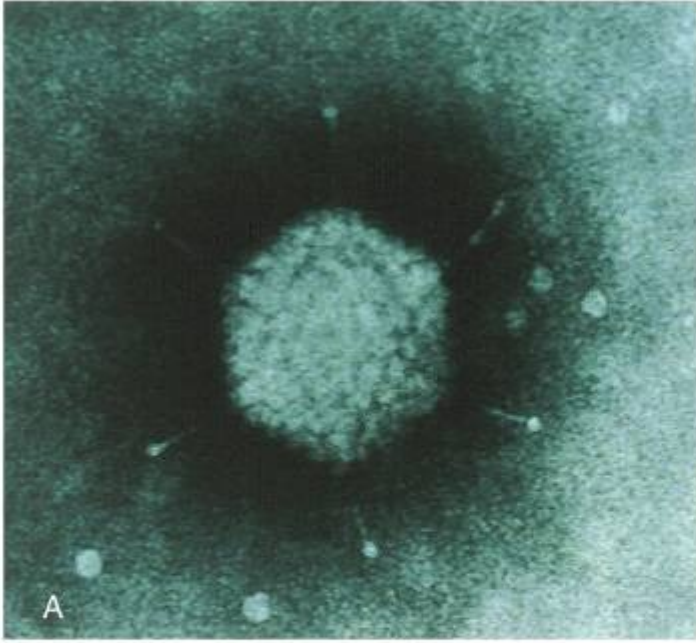
Adenoviruses

Classification & structure

- Reflects the recovery of the initial isolate from explants of human adenoids
- Family: Adenoviridae
- Genera
 - 1 Aviadenovirus – Infect birds
 - 2 Mastadenovirus – Infect mammals
- **Linear ds** DNA Genome
- **70-90 nm in size**
- Icosahedral capsid with 252 capsomeres

Cont...

- Non – enveloped
- More than 49 human serotypes are known
- Common serotypes: 1 - 8, 11, 21, 35, 37, 40 & 41
- ▶ Types 40 & 41 are enteric pathogens



Adenovirus Structure

Cont...

General Properties

Not easily affected by:

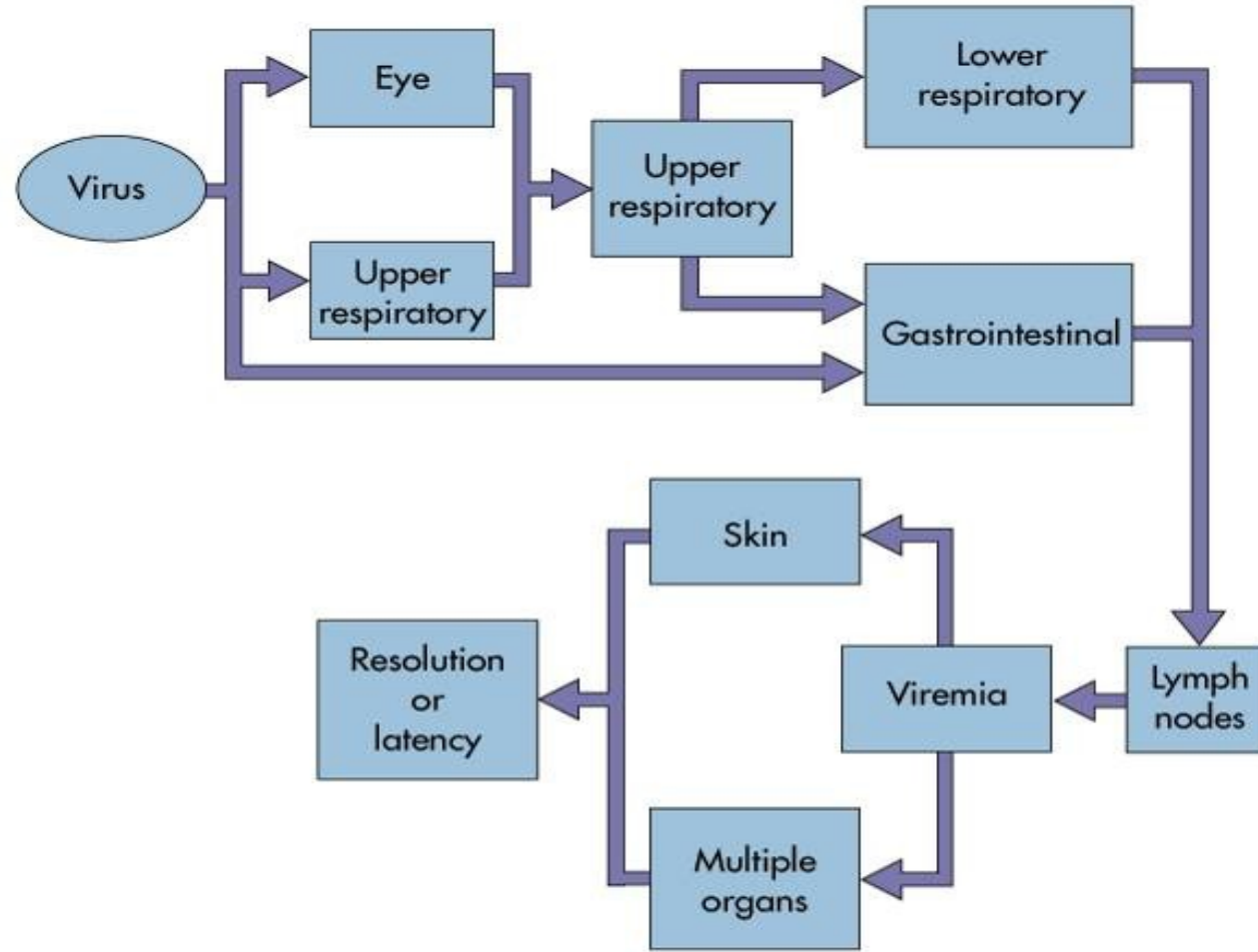
- External environment
- Low PH
- Bile salts & proteolytic enzymes
- Can replicate to high titer in the gastro intestinal tract

Cont...

Pathogenesis

- Infect and replicate in epithelial cells of respiratory tract, eye, gastrointestinal tract, urinary bladder, and liver.
- Usually do not spread beyond the regional lymph nodes
- Group C viruses persist as latent infections for years in adenoids and tonsils
 - shed in the feces for many months after the initial infection

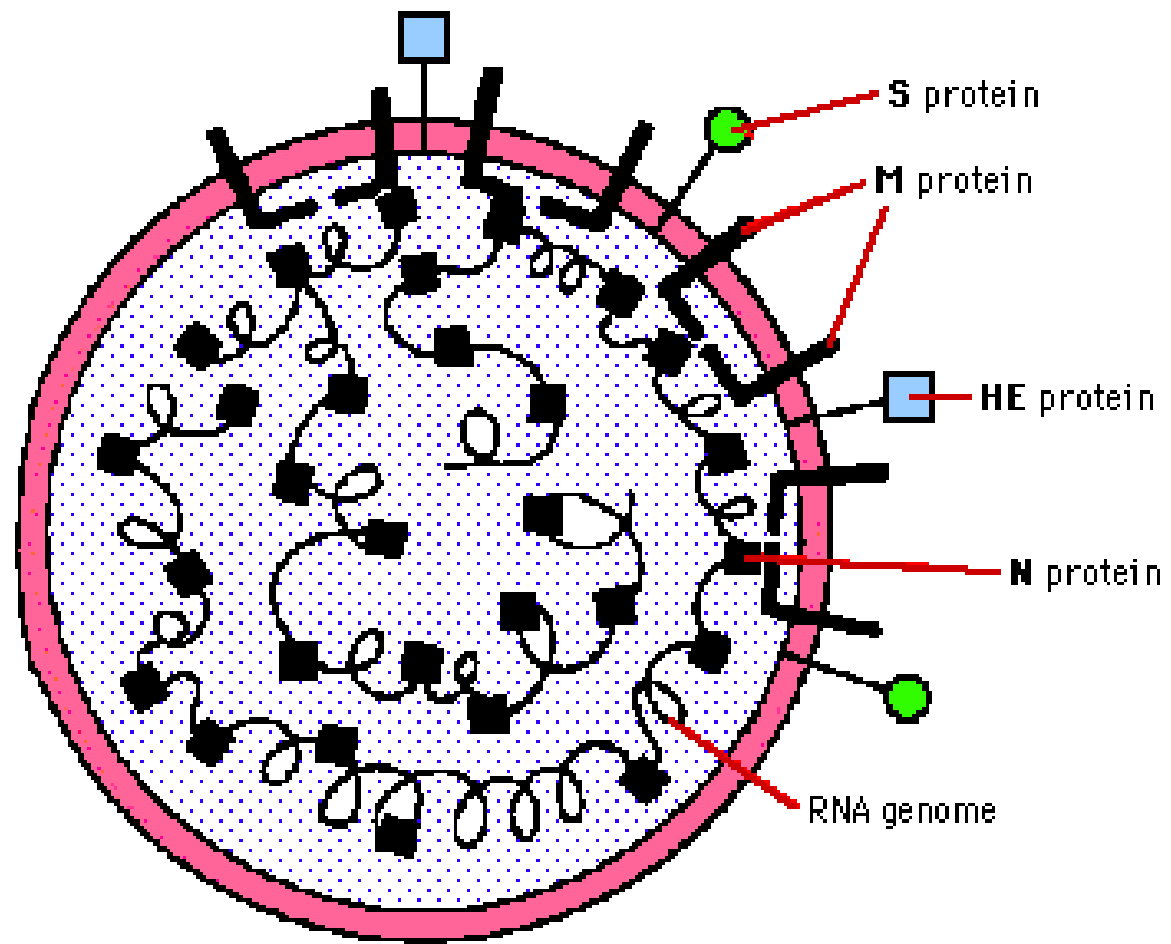
Adenovirus pathogenesis



Coronaviruses

Properties

- Spherical/ irregularly-shaped
- 80-220 nm diameter
- Envelope bearing distinctive, 'club-shaped' peplomers - 'crown-like' appearance
 - three glycoproteins:
 - **S** - Spike protein: receptor binding, cell fusion, major antigen
 - **M** - Membrane protein: transmembrane - budding & envelope formation
 - **HE** – Hemagglutinin - hemagglutination, acetylsterase activity
- ss RNA (20-30kb), linear, non segmented, positive sense
 - the largest genome among RNA viruses



Coronavirus

Cont...

Classification

- *Family → Coronaviridae*
- *Genera → 1 Coronavirus - Man, animals and birds*
2 Torovirus – Ungulates- associated with diarrhea
- Two antigenic groups of human coronaviruses
 - HCV 229 E and HCV OC 43
 - Coronaviruses of domestic animals and rodents are included in these two groups
- Third group → the Avian infectious bronchitis virus of chickens
- Cross reactions occur between some human and some animal strains

Cont...

Pathogenesis

- ❑ Highly species-specific
- ❑ Little is known
- ❑ Display a tropism for epithelial cells of the respiratory or gastrointestinal tract
- ❑ Remain limited to the upper respiratory tract
 - a slow, patchy destruction of ciliated epithelial cells and the loss of beating cilia
 - disease development

Parainfluenzaviruses

Properties, classification and Structure

❖ Paramyxoviridae

❖ Morphology

- resembles that of influenza viruses
- larger and much more pleomorphic
- enveloped
- linear, single-stranded RNA, and negative sense
- non-segmented
 - antigenically stable

Cont...

- Four serotypes in human 1-4
- No common group antigen
- Closely related to Mumps virus
- Types 1,2,3 severe second only to RSV
- Type 4 mild URT
- Wide distribution
- Direct person-to-person or large-droplet aerosols

Cont...

Pathogenesis

- Replication
 - respiratory epithelia
- Viremia is uncommon
- Involve only the nose and throat
 - a harmless “common cold” syndrome
- More extensive and, especially with types 1 & 2, involve the larynx and upper trachea
 - croup (laryngotracheobronchitis)
 - respiratory obstruction due to swelling of the larynx and related structures

Cont...

- Spread deeper to the lower trachea and bronchi
 - pneumonia or bronchiolitis (or both) (type 3)
- Host cell protease
 - cleave & activate the fusion protein of virus
 - replicate well & disseminate throughout the respiratory tract

Influenza viruses

Structure and Properties

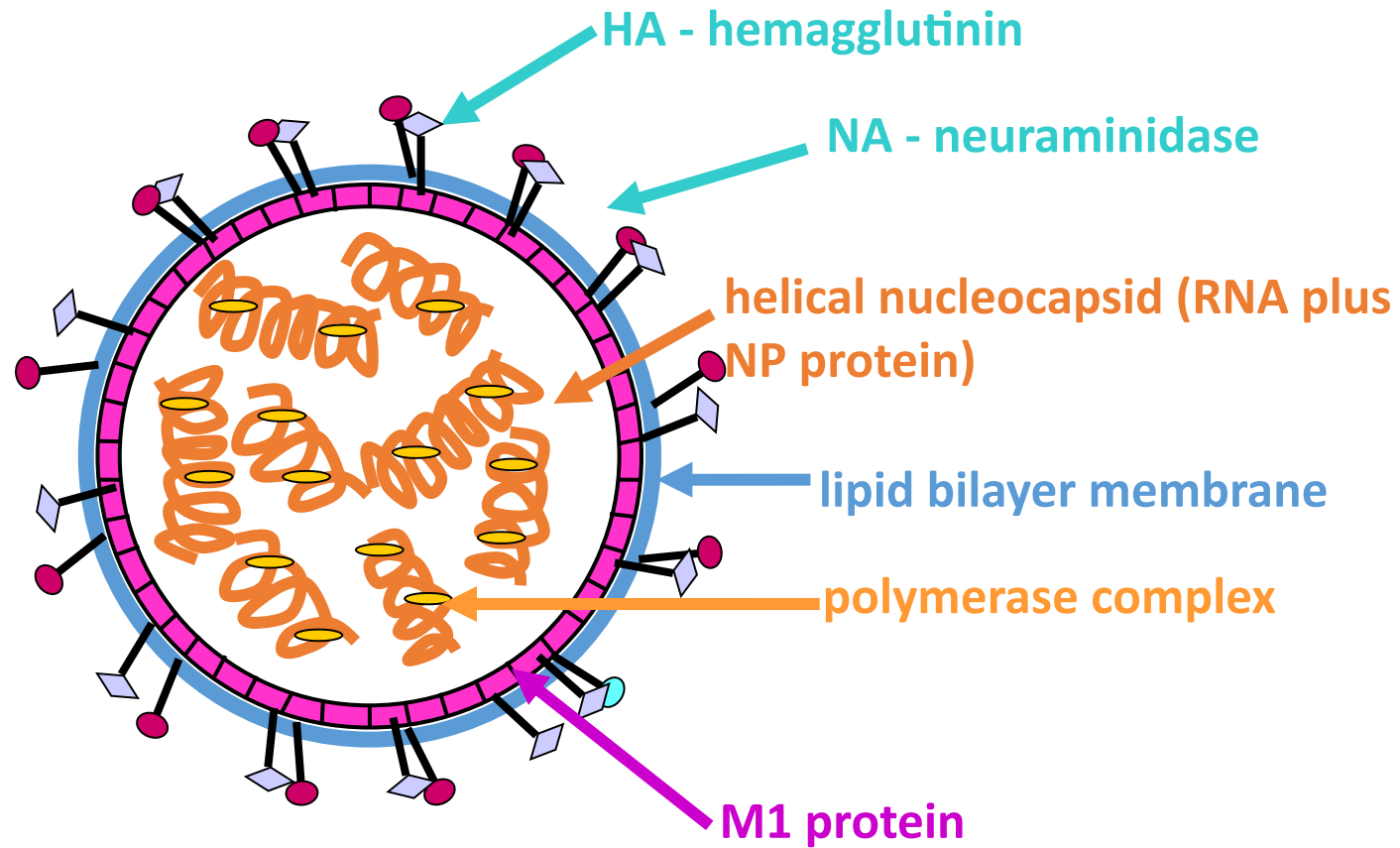
- All known orthomyxoviruses are influenza viruses
- Spherical, Pleomorphic
- 80-120 nm
- The single stranded RNA genomes of Influenza A and B occur as 8 separate segments
- Influenza C viruses contain 7 segments of RNA lacking a neuraminidase gene

Cont...

- Nine different structural proteins
- Nucleoprotein (NP) + viral RNA → a ribonucleoprotein (RNP) structure
- Three large proteins (PB1, PB2 and PA) + viral RNP
 - RNA transcription and translation
- Matrix protein
 - particle morphogenesis
 - a major component of the virion (40% of viral protein)

Cont...

- A lipid envelope surrounds the virus particle
- Two glycoproteins (hemagglutinin (HA) & neuraminidase (NA))
 - envelope
 - exposed as spikes
 - determine antigenic variation of influenza viruses and host immunity
 - undergo antigenic variation independent of each other



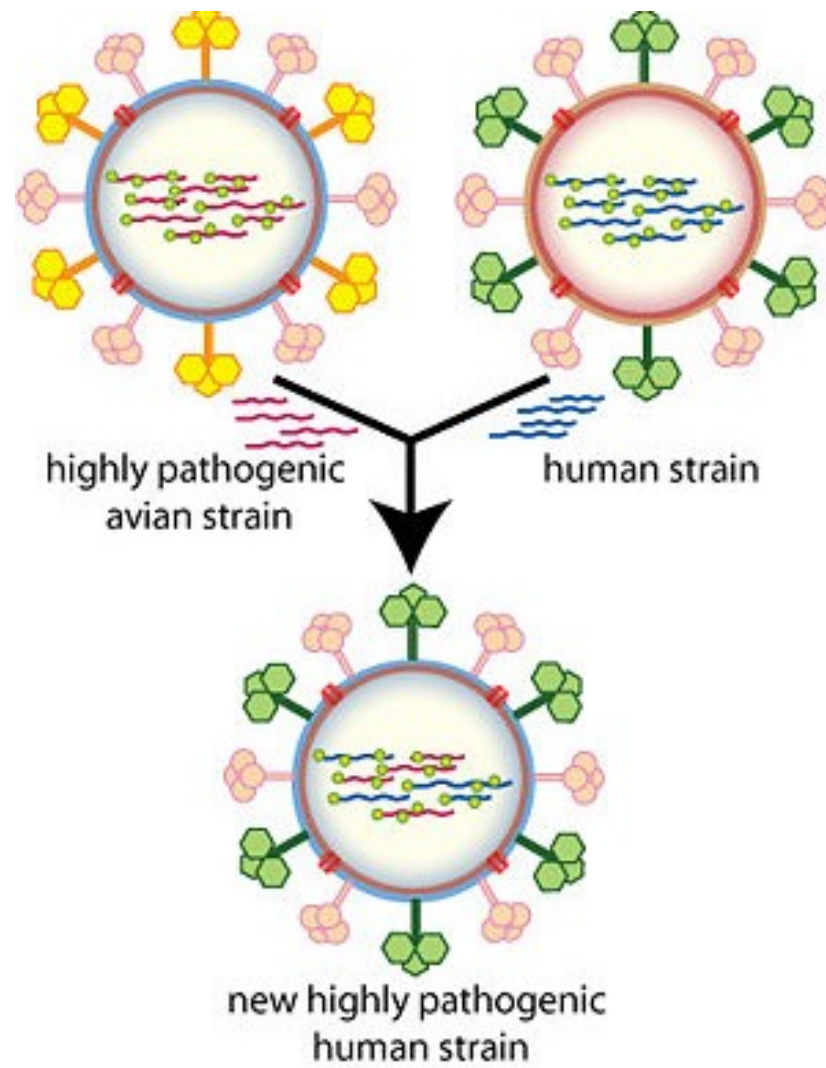
Influenza viruses

Cont...

- Minor antigenic changes → antigenic drift
 - accumulation of point mutations in the gene
 - amino acid changes in the protein
 - alter antigenic sites on the molecule
 - a virion can escape recognition by the host's immune system

Cont...

- Major antigenic changes in HA or NA → antigenic shift
 - a new subtype
 - segmented nature of the genome
 - cell is coinfecting by 2 different viruses of a given type
 - mixtures of parental gene segments may be assembled in to progeny virions
 - genetic reassortment → sudden changes in viral surface antigens



Antigenic shift

Cont...

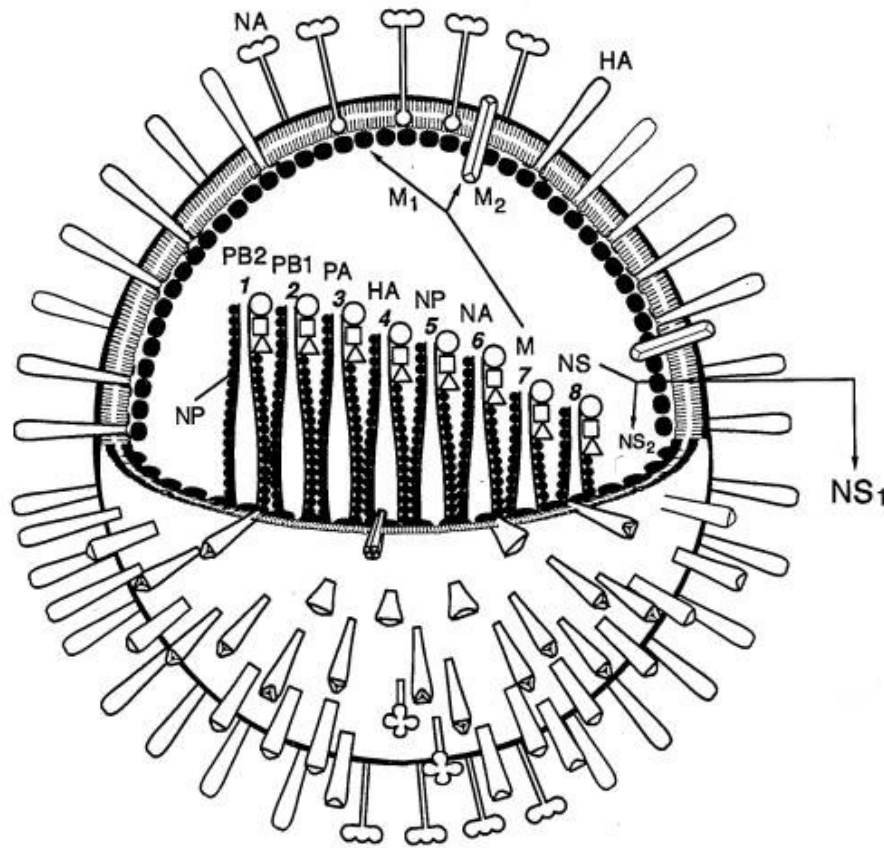
- Explains the epidemiological features of influenza
- Poses significant problems for vaccine development
- Influenza B and C viruses do not exhibit antigenic shift
 - few related viruses exist in animals

Table 32.1 Major Antigenic Shifts Associated with Influenza A
Pandemics, 1947–1987

Year	Subtype	Prototype Strain
1947	H ₁ N ₁	A/FM ₁ /47
1957	H ₂ N ₂	A/Singapore/57
1968	H ₃ N ₂	A/Hong Kong/68
1977	H ₁ N ₁	A/USSR/77
1987	H ₃ N ₂	No pandemic occurred; various strains circulating worldwide

Cont...

- Pandemic
 - antigenic shift
 - absence of natural immunity
- Replication in nucleus
- RNA segments encode one or more protein



Structure of influenza virus. The diagram illustrates the main structural features of the virion. The surface of the particle contains three kinds of spike proteins: the hemagglutinin (HA), neuraminidase (NA), and matrix (M2) protein embedded in a lipid bilayer derived from the host cell and covers the matrix (M1) protein that surrounds the viral core. The ribonucleoprotein complex making up the core consists of at least one of each of the eight single-stranded RNA segments associated with the nucleoprotein (NP) and the three polymerase proteins (PB2, PB1, PA). RNA segments have base pairing between their 3' and 5' ends forming a panhandle. Their organization and the role of NS2 in the virion remain unresolved. (From Fields Virology, 4th ed, Knipe & Howley, eds, Lippincott Williams & Wilkins, 2001, Fig. 47-2)

Influenza gene functions

Segment*	Protein	Function
1	PB2	Polymerase component
2	PB1	Polymerase component
3	PA	Polymerase component
4	HA	Hemagglutinin, viral attachment protein, fusion protein, target of neutralizing antibody
5	NP	Nucleocapsid
6	NA	Neuraminidase (cleaves sialic acid and promotes virus release)
7 [†]	M ₁	Matrix protein: Viral structural protein (interacts with nucleocapsid and envelope, promotes assembly)
	M ₂	Membrane protein (forms membrane channel and target for amantadine, facilitates uncoating and HA production)
8 [†]	NS ₁	Nonstructural protein (inhibits cellular messenger RNA translation)
	NS ₂	Nonstructural protein (important but unknown function)

*Listed in decreasing order of size.

[†]Encodes two messenger RNAs.

Cont...

Classification and Nomenclature

- ✓ Orthomyxoviridae
- ✓ Genus
 - Influenzavirus A, B
 - human and animal strains of Influenza type A and human strains of type B
 - Influenzavirus C
 - Influenza type C viruses of humans and swine
- ✓ Nucleoprotein (NP) and matrix protein (M)
 - Influenza viruses into types A, B and C
 - possess no cross-reactivity among the 3 types

Cont...

- ✓ Hemagglutinin (HA) and neuraminidase (NA)
 - subtype the viruses
- ✓ 15 subtypes of HA (H1-H15) and 9 subtypes of NA (N1-N9)
 - birds, animals or humans
- ✓ 4 HA (H1-H3, H5) and 2 NA (N1, N2) subtypes
 - humans

Cont...

- ✓ Nomenclature system
 - type
 - host of origin
 - geographic origin
 - strain number
 - year of isolation

Cont...

- antigenic descriptions of the HA and the NA are given in parenthesis for type A

- the host of origin is not indicated for human isolates

e.g., A/Hong Kong/03/68(H3 N2)

- but it is for others

e.g., A/Swine/Iowa/15/30(H1 N1)

Cont...

Pathogenesis

- A few cells of respiratory epithelium are infected
- Progeny virions are soon produced and spread to adjacent cells
 - replicative cycle is repeated
- Neuraminidase lowers the viscosity of the mucus film in the respiratory tract
 - laying bare the cellular surface receptors
 - promoting the spread of virus-containing fluid to lower portions of the tract
- Many cells in the respiratory tract are infected and eventually killed

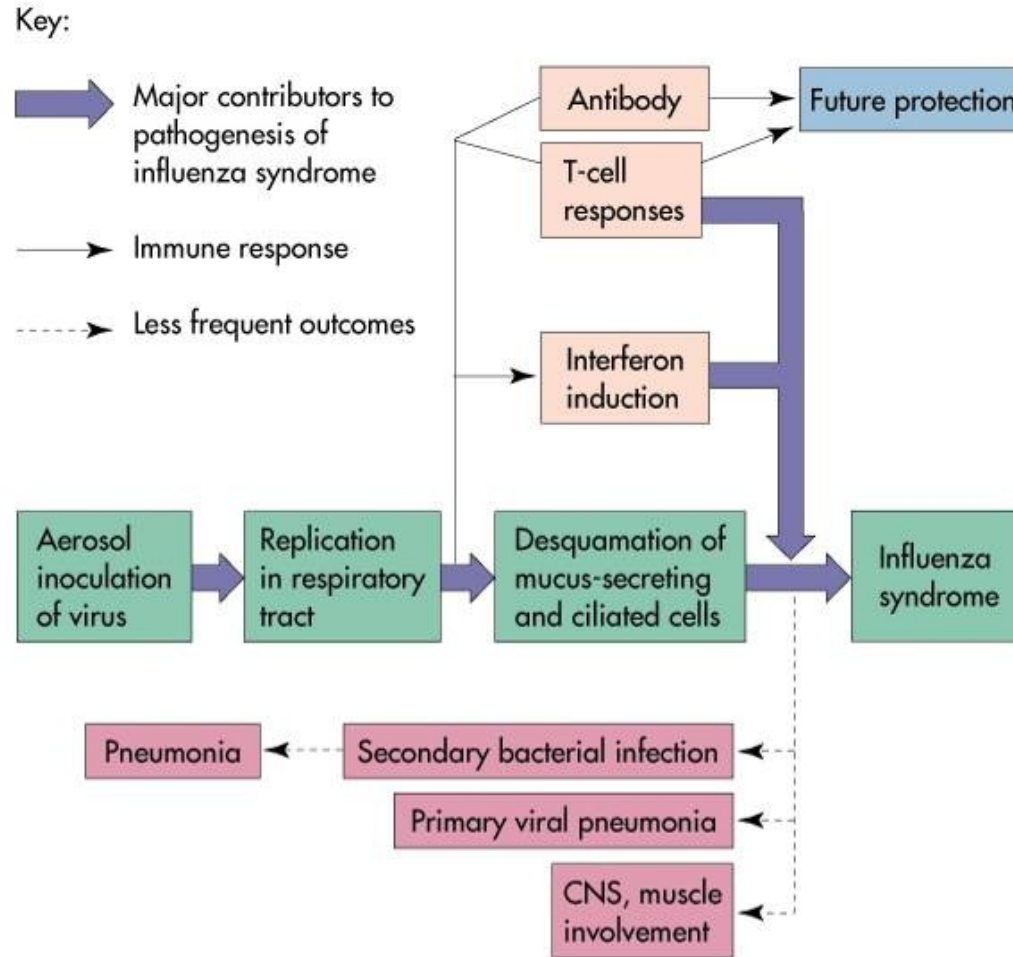
Cont...

- Cellular destruction and desquamation of superficial mucosa of the respiratory tract
- Do not affect the basal layer of epithelium
- Edema and mononuclear infiltrations
 - local symptoms
- The prominent systemic symptoms
 - production of cytokines

Cont...

- Interferon response
 - host recovery from infection
- Viral damage to the respiratory tract epithelium lower its resistance to secondary bacterial invaders
 - *S. pneumoniae*, *S. aureus* and *Hemophilus influenzae*

Influenza pathogenesis



Respiratory syncytial virus

Properties, classification and Structure

❖ Paramyxoviridae

❖ Morphology

- resembles that of influenza viruses
- larger and much more pleomorphic
- enveloped
- linear, single-stranded RNA, and negative sense
- non-segmented
 - antigenically stable

Cont...

Pathogenesis

- Replication occurs initially in epithelial cells of the nasopharynx
- Spread into the lower respiratory tract, probably carried there by secretions
- Viremia has not been detected
- An intact immune system seems to be important in resolving an infection
- Spread outside the respiratory epithelium (kidney, liver, etc)
 - in several fatal infections in individuals who lacked cell-mediated immunity

Group A streptococcus (*S. pyogenes*)

- ❖ Gram-positive cocci
- ❖ Pairs or chains
- ❖ Facultative anaerobs
- ❖ Some grow only in an atmosphere enhanced with carbon dioxide (capnophilic growth)
- ❖ Nutritional requirements are complex
 - blood- or serum-enriched media for isolation
- ❖ Carbohydrates are fermented
 - lactic acid
- ❖ Catalase negative

Cont...

Pathogenesis

➤ Virulence

- ability of the bacteria to adhere to the surface of host cells
- invade into the epithelial cells
- avoid opsonization and phagocytosis
- produce a variety of toxins and enzymes

➤ More than 10 different bacterial antigens

- mediate adherence to host cells
- lipoteichoic acid, M protein, and F protein
 - most important

Cont...

- The initial adherence is a weak interaction between lipoteichoic acid and fatty acid binding sites on fibronectin and epithelial cells
- Subsequent adherence
 - M protein, F protein, and other adhesins
 - specific host cell receptors
- Invade into epithelial cells
 - M protein and F protein, as well as other bacterial antigens
 - maintenance of persistent infections
 - invasion into deep tissue

Cont...

- Has multiple mechanisms for avoiding opsonization and phagocytosis
- The conserved region of M protein can bind the serum β -globulin, factor H, which is a regulatory protein for the alternative complement pathway
- C3b, an important mediator of phagocytosis, is destabilized by factor H.
- When C3b binds to the cell surface in the region of the M protein, it is degraded by factor H, and phagocytosis is prevented

Cont...

- The binding of fibrinogen to the cell surface of M protein also blocks activation of complement by the alternate pathway and reduces the amount of bound C3b
- The M-related proteins interfere with phagocytosis
- Produce C5a peptidase
 - inactivate C5a
 - block chemotaxis of neutrophils and mononuclear phagocytes
- A number of enzymes and toxins can mediate the pathology

Cont...

Pyrogenic Exotoxins (Spes)

- ✓ Erythrogenic toxins
- ✓ Three toxins (SpeA, SpeB, and SpeC)
- ✓ Superantigens
- ✓ Interact with both macrophages and helper T cells
- ✓ Cytokines
 - (1) interleukin-1 (IL-1), IL-2, and IL-6
 - (2) tumor necrosis factor- α (TNF- α) and TNF- β
 - (3) interferon- γ

Cont...

→ mediate a variety of important effect

- the shock and the organ failure

◆ streptococcal toxic shock syndrome

✓ Rash observed in patients with scarlet fever

Cont...

Streptolysins S and O

Streptolysin S

- An oxygen-stable, nonimmunogenic, cell-bound hemolysin
- Lyse erythrocytes, leukocytes, and platelets
- release of lysosomal content after engulfment, with subsequent death of the phagocytic cell
- Produced in the presence of serum
- β -hemolysis seen on blood agar media

Cont...

Streptolysin O

- An oxygen-labile hemolysin
- Lysing erythrocytes, leukocytes, platelets, and cultured cells
- Antibodies are readily formed against streptolysin O
 - documenting recent *S. pyogen* infection (ASO test)

Streptokinase

- Two forms of streptokinase
- Lyse blood clots
- Rapid spread of *S.pyogens* in infected tissues

Cont...

Deoxyrinonuclease

- ❑ Four deoxyribonucleases (Dnases A to D)
- ❑ Are not cytolytic
- ❑ Depolymerize free DNA present in pus
- ❑ Reduces the viscosity of the abscess material
- ❑ Facilitates spread of the organisms

Corynebacterium diphtheriae

- An irregularly staining, pleomorphic, gram-positive bacillus
- Aerobic or facultatively anaerobic, nonmotile and catalase positive
- Metachromatic granules
 - methylene blue

Pathogenesis

- ✓ A-B exotoxin (diphtheria toxin)
- ✓ Three functional regions
 - a **receptor binding region** and a **translocation region** on the B subunit
 - a **catalytic region** on the A subunit

Cont...

✓ Receptor

→ heparin-binding epidermal growth factor

- surface of many eukaryotic cells

→ heart and nerve cells

→ cardiac and neurologic symptoms observed in patients with severe diphtheria

✓ The translocation region is inserted into the endosomal membrane

→ movement of the catalytic region into the cytosol

✓ The A subunit then terminates host cell protein synthesis by inactivating elongation factor 2 (EF-2)

Neisseria gonorrhoeae

- ❖ Aerobic, gram-negative cocci typically arranged in pairs (diplococci)
- ❖ Not motile and do not form endospores
- ❖ Oxidase-positive
- ❖ Most produce catalase

Cont...

Pathogenesis

- Attach to mucosal cells, penetrate into the cells and multiply
- Pass through the cells into the subepithelial space, where infection is established
- pili → initial attachment
- Opa protein directs first a tighter association with the host cell surface and then migration of bacteria
- The Por protein protects the phagocytosed bacteria from intracellular killing by inhibiting phagolysosome fusion
- Lipooligosaccharide (LOS)
 - inflammatory response and release of TNF- α
 - symptoms associated with gonococcal disease

Cont...

GENERAL DIAGNOSTIC APPROACHES

- Viruses
 - the vast majority of upper respiratory infections
 - not amenable to specific therapy
 - laboratory tests are usually reserved for investigating outbreaks or in cases in which the illness seems unusually severe or atypical
- The primary diagnostic approach in pharyngitis and tonsillitis is to determine whether there is a bacterial cause requiring specific treatment

Cont...

- The only reliable method is to collect a throat swab for culture
 - made only to detect the presence or absence of group A streptococci
- Direct antigen tests for rapidly detecting *S. pyogenes* in throat swabs
 - enzyme immunoassay or latex agglutination–based methods
 - lack of sensitivity

Cont...

- For the laboratory diagnosis of diphtheria or pharyngeal gonorrhea, the clinical suspicion should be indicated to the laboratory so that specific cultures for *C. diphtheriae* or *N. gonorrhoeae* may be made

GENERAL PRICIPLES OF MANAGEMENT

- Viral infections of the upper respiratory tract can only be treated symptomatically
- *S. pyogenes*
 - penicillin
 - patient is allergic to penicillin
 - erythromycin or a cephalosporin

Cont...

- *C. diphtheriae* infections
 - more complex management
 - Early administration of diphtheria antitoxin to specifically neutralize the exotoxin before it is bound by the host cell
 - Antibiotic therapy with penicillin or erythromycin
 - eliminate *C. diphtheriae* and terminate toxin production
- ceftriaxone, cefixim, ciprofloxacin, or ofloxacin be used as initial therapy for infections caused by *N. gonorrhoeae*

Middle and Lower Respiratory Infections

MIDDLE RESPIRATORY TRACT INFECTION

- Comprise the epiglottis, surrounding aryepiglottic tissues, larynx, trachea, and bronchi
- Inflammatory disease involving these sites may be localized (eg, laryngitis) or more widespread (eg, laryngotracheobronchitis)
- The majority of severe infections occur in infancy and childhood

Cont...

CLINICAL FEATURES

Epiglottitis

- abrupt onset of throat and neck pain, fever, and inspiratory stridor
- phonation becomes difficult
- difficulty in swallowing
- death may result from acute airway obstruction

Cont...

Laryngitis or its more severe form, **croup**

- an abrupt onset (spasmodic croup) or develop more slowly over hours or a few days as a result of spread of infection from the upper respiratory tract
- variable fever; inspiratory stridor; hoarse phonation; and a harsh, barking cough
- the inflammation is localized to the subglottic laryngeal structures, including the vocal cords
- It sometimes extends to the trachea (laryngotracheitis) and bronchi (laryngotracheobronchitis)
 - a deeper, more severe cough that may provoke chest pain

Cont...

- transient aphonia

Bronchitis or tracheobronchitis

- primary manifestation of infection or a result of spread from upper respiratory tissues
- cough, variable fever, and sputum production, which is often clear at the onset but may become purulent as the illness persists

Chronic bronchitis

- long-standing damage to the bronchial epithelium
- cigarette smoking

Cont...

COMMON ETIOLOGIC AGENTS

- ❑ With the exception of epiglottitis, acute diseases of the middle airway are usually caused by viral agents
- ❑ When acute airway obstruction is present, noninfectious possibilities, such as aspirated foreign bodies and acute laryngospasm or bronchospasm caused by anaphylaxis, must also be considered

Major Causes of Acute Middle Respiratory Tract Disease

SYNDROME	VIRUSES	BACTERIA	PERCENTAGE CAUSED BY VIRUSES
Epiglottitis	Rare	<i>H. influenzae</i> , <i>S. pneumoniae</i> , <i>C. diphtheriae</i> , <i>N. Meningitidis</i>	10
Laryngitis and croup	Parainfluenza viruses, influenza viruses, adenoviruses; occasionally respiratory syncytial virus, rhinoviruses, coronaviruses, echoviruses	Rare	90
Laryngotracheitis and laryngotracheobronchitis	Same as for laryngitis and croup	<i>H. influenzae</i> , <i>Staphylococcus aureus</i>	90
Bronchitis	Parainfluenza viruses, influenza viruses, respiratory syncytial virus, adenoviruses, measles	<i>Bordetella pertussis</i> , <i>H. influenzae</i> , <i>M. pneumoniae</i> , <i>C. pneumoniae</i>	80

Haemophilus influenzae

- *Haemophilus*
- small, sometimes pleomorphic, gram-negative rods
- Obligate parasites
- *Haemophilus influenzae*
 - disease, with infections most often reported in pediatric patients before the introduction of the H. influenzae type b (HIB) vaccine

Cont...

Physiology and Structure

- ❑ requires supplementation of media with one or both of the following growth-stimulating factors
 - (1) hemin (also called X factor for unknown factor)
 - (2) nicotinamide adenine dinucleotide (NAD; also called V factor for "vitamin")
- ❑ both factors are present in blood-enriched media
- ❑ sheep blood agar must be gently heated to destroy the inhibitors of V factor
 - heated blood ("chocolate") agar is used for the in vitro isolation of *Haemophilus*

Cont...

- ❑ The cell wall structure
 - typical of other gram-negative rods
- ❑ Lipopolysaccharide with endotoxin activity
 - cell wall
- ❑ strain-specific and species-specific proteins
 - outer membrane
- ❑ The surface of many but not all strains of *H. influenzae* is covered with a polysaccharide capsule
 - six antigenic serotypes (a through f)

Cont...

- ❑ Before the introduction of the Hib vaccine, *H. influenzae* serotype b
 - more than 95% of all invasive Haemophilus infections
- ❑ After the introduction of the vaccine, most disease
 - disappeared
- ❑ more than half of all invasive disease is now caused by nonencapsulated (nontypeable) strains
- ❑ subdivided into eight biotypes (I through VIII)
 - indole production, urease activity, and ornithine decarboxylase activity
 - useful for epidemiologic purposes

Cont...

Pathogenesis

- ❖ Pili and nonpilus adhesins
 - colonization of the oropharynx with *H. influenzae*
- ❖ Cell wall components of the bacteria impair ciliary function
 - damage of the respiratory epithelium
- ❖ Translocated across both epithelial and endothelial cells
- ❖ Blood stream
- ❖ High- grade bacteremia can develop, with dissemination to the meninges or other distal foci

Cont...

❖ Major virulence factor

- antiphagocytic polysaccharide capsul

 - ribose, ribitol, and phosphate (polyribitol phosphate [PRP])

❖ Antibodies directed against the capsule greatly stimulate bacterial phagocytosis and complement-mediated bactericidal activity

- natural infection, vaccination with purified PRP, or the passive transfer of maternal antibodies

❖ The risk of meningitis and epiglottitis is significantly greater in patients with no anti-PRP antibodies, those with depletion of complement, and those who have undergone splenectomy

Cont...

- ❖ The lipopolysaccharide lipid A component induces meningeal inflammation in an animal model
 - initiating this response in humans
- ❖ Immunoglobulin IgA1 proteases are produced by *H. influenzae*
 - colonization of the organisms on mucosal surfaces by interfering with humoral immunity

Streptococcus pneumoniae

- ❖ Gram-positive cocci
- ❖ Pairs or chains
- ❖ Facultative anaerobs
- ❖ Some grow only in an atmosphere enhanced with carbon dioxide (capnophilic growth)
- ❖ Nutritional requirements are complex
 - blood- or serum-enriched media for isolation

Cont...

❖ Carbohydrates are fermented
→ lactic acid

❖ Catalase negative

Pathogenesis

Colonization and migration

- Colonizes the oropharynx
- Spread to the lungs, paranasal sinuses, or middle ear
- Transported in the blood stream to distal sites (brain)

Cont...

- Initial colonization of the oropharynx
 - binding of the bacteria to epithelial cells by means of **surface protein adhesins**
- Secretory IgA protease and pneumolysin
- Secretory IgA traps bacteria in mucin by attaching itself to the bacteria at the antigen-binding site and to mucin at the Fc region
- Secretory IgA protease prevents this interaction
- Pneumolysin binds cholesterol in the host cell membrane and creates pores
 - destroy the ciliated epithelial cells as well as phagocytic cells
- Subsequent migration

Cont...

Tissue Destruction

- ❑ Mobilization of inflammatory cells to the focus of infection
- ❑ Pneumococcal teichoic acid, peptidoglycan fragments and pneumolysin
- ❑ Teichoic acid and the peptidoglycan fragments
 - alternative complement pathway
 - C5a → inflammatory process

Cont...

❑ Pneumolysin

→ classic complement pathway

- C3a, and C5a

→ activated leukocytes

- cytokines (IL-1 & TNF- α)

→ further migration of inflammatory cells to the site of infection, fever, tissue damage, and other signs characteristic of pneumococcal infection

❑ Hydrogen peroxide

→ tissue damage caused by reactive oxygen intermediates

Cont...

Phagocytic survival

- ❖ Survives phagocytosis
- ❖ Anti-phagocytic protection afforded by its capsule
- ❖ Pneumolysin-mediated suppression of the phagocytic cell oxidative burst
 - intracellular killing
- ❖ Virulence
 - capsule

Neisseria meningitidis

- ❖ Aerobic, gram-negative cocci typically arranged in pairs (diplococci)
- ❖ Not motile and do not form endospores
- ❖ Oxidase-positive
- ❖ Most produce catalase

Pathogenesis

- attach selectively to specific receptors for meningococcal pili on nonciliated columnar cells of the nasopharynx

Cont...

- Internalized into phagocytic vacuoles
- Avoid intracellular death, replicate and then migrate to the subepithelial space
- Capsule protect *N. meningitidis* from phagocytic destruction
- The diffuse vascular damage (e.g., endothelial damage, inflammation of vessel walls, thrombosis, disseminated intravascular coagulation)
 - LOS endotoxin

Staphylococcus aureus

Features of staphylococci

- Gram-positive cocci
- Grow in a pattern resembling a cluster of grapes
- Single cells, pairs, or short chains
- 0.5 to 1µm in diameter
- Nonmotile, aerobic or facultative anaerobic
- Catalase positive
- Grow in medium containing 10% sodium chloride
- Temperature ranging from 18°C to 40°C
- present on the skin and mucous membranes of humans, other mammals, and birds

Cont...

- 32 species and 15 subspecies
 - found on humans
- Most commonly associated with human diseases
 - *S. aureus*, *S. epidermidis*, *S. saprophyticus*, *S. capitis* and *S. haemolyticus*
- *S. aureus*
 - most virulent and best-known
 - produces the enzyme coagulase
- All other species
 - coagulase negative staphylococci

Cont...

Pathogenesis

Staphylococcal toxins

○ *S. aureus*

→ many virulence factors

- at least five cytolytic or membrane damaging toxins (alpha, beta, delta, gamma, and Panton-Valentine [P-V] leukocidin)
- toxic shock syndrome toxin-1 (TSST-1)

Cont...

Alpha Toxin

- Disrupts the smooth muscle in blood vessels
- Toxic to many types of cells
 - erythrocytes, leukocytes, hepatocytes, platelets, and cultivated cells (e.g., human diploid fibroblasts, HeLa cells, Ehrlich ascites carcinoma cells)
 - integrated in the hydrophobic regions of host cell membrane
 - formation of 1- to 2nm pores
 - the rapid efflux of K^+ and influx of Na^+ , Ca^{2+} , and other small molecules
 - osmotic swelling and cell lysis

Cont...

→ mediator of tissue damage in staphylococcal disease

Beta Toxin

- ❑ Sphingomyelinase C

- ❑ specificity for sphingomyelin and lysophosphatidylcholine

- ❑ Toxic to a variety of cells

 - erythrocytes, leukocytes, macrophages, and fibroblasts

- ❑ hydrolysis of membrane phospholipids in susceptible cells, with lysis proportional to the concentration of sphingomyelin exposed on the cell surface

 - differences in species susceptibility to the toxin

Cont...

❑ Together with alpha toxin

→ tissue destruction and abscess formation

Delta Toxin

- ❖ A wide spectrum of cytolytic activity, affecting erythrocytes, many other mammalian cells, as well as intracellular membrane structures
- ❖ A surfactant disrupting cellular membranes by means of a detergent like action

Cont...

Gamma Toxin and Panton-Valentine Leukocidin

- Lyse neutrophils and macrophages
- Cell lysis is mediated by pore formation with subsequent increased permeability to cations and osmotic instability

Toxic Shock Syndrome Toxin-1 (TSST-1)

- Virtually all *S. aureus* strains responsible for menstruation associated toxic shock syndrome (TSS) and half of the strains responsible for other forms of TSS produce TSST-1
- Superantigen
 - nonspecific cytokine release from macrophages and T lymphocytes

Cont...

- leakage of endothelial cells at low concentrations and a cytotoxic effect to the cells at high concentrations
- Ability of TSST-1 to penetrate mucosal barriers
 - systemic effects of TSS
- Death in patients with TSS
 - hypovolemic shock
 - multiorgan failure

Cont...

Staphylococcal Enzymes

Coagulase

- Two forms of coagulase
 - bound and free
- Coagulase bound to the staphylococcal cell wall
 - fibrinogen to insoluble fibrin
 - staphylococci to clump

Cont...

- The cell free coagulase
 - reacting with a globulin plasma factor (coagulase-reacting factor) to form staphylothrombin, a thrombin-like factor
 - fibrinogen to insoluble fibrin
- The role of coagulase in the pathogenesis of disease is speculative
- Cause the formation of a fibrin layer around a staphylococcal abscess
- Localizing the infection and protecting the organisms from phagocytosis

Cont...

Catalase

- ❑ all staphylococci produce catalase
- ❑ Toxic hydrogen peroxide to water and oxygen
- ❑ Hydrogen peroxide can accumulate during bacterial metabolism or after phagocytosis

Hyaluronidase

- ❖ Hydrolyzes hyaluronic acids, the acidic mucopolysaccharides present in theacellular matrix of connective tissue
- ❖ Facilitates the spread of *S. aureus* in tissues

Cont...

Fibrinolysin

- Staphylokinase
- Dissolve fibrin clots
- Distinct from the fibrinolytic enzymes produced by streptococci

Penicillinase

- More than 90% of staphylococcal isolates were susceptible to penicillin in 1941, the year the antibiotic was first used clinically
- Resistance to penicillin quickly developed
 - produce penicillinase (β -lactamase)
- The widespread distribution of this enzyme was ensured by its presence on transmissible plasmids

Bordetella pertussis

- Extremely small, strictly aerobic, nonfermentative, gram-negative coccobacilli
- nonmotile and oxidize amino acids
- do not ferment carbohydrates

Pathogenesis

- Infection with *B. pertussis* and the development of whooping cough require exposure to the organism, bacterial attachment to the ciliated epithelial cells of the respiratory tract, proliferation of the bacteria, and production of localized tissue damage and systemic toxicity

Cont...

- Pertactin and filamentous hemagglutinin contain an Arg-Gly-Asp sequence (RGD motif)
 - binding to sulfated glycoprotein integrins on the membranes of ciliated respiratory cells
 - bind to CR3, a glycoprotein receptor on the surface of macrophages
 - the phagocytosis of the bacteria without initiating an oxidative burst, which is important in the intracellular survival and replication of the bacteria
 - protects *B. pertussis* from humoral antibodies

Cont...

- Pertussis toxin
 - A-B toxin
 - a toxic subunit (S1) and five binding subunits (S2 to S5; two S4 subunits are present in each toxin molecule)
- The S2 subunit binds to lactosylceramide, a glycolipid present on ciliated respiratory cells
- The S3 subunit binds to receptors on phagocytic cells
 - an increase in CR3 on the cell surface, which facilitates attachment mediated by the pertactin and filamentous hemagglutinin and subsequent bacterial phagocytosis

Cont...

- several toxins that mediate the localized and systemic manifestations of disease
- S1 portion
 - adenosine diphosphate (ADP)-ribosylating activity for the membrane surface G proteins
 - regulate adenylate cyclase activity
- Pertussis toxin inactivates $G_i\alpha$, the inhibitory protein that controls adenylate cyclase activity
- The uncontrolled expression of the enzyme
 - an increase in cyclic adenosine monophosphate (cAMP) levels and a subsequent increase in respiratory secretions and mucus production characteristic of the paroxysmal stage of pertussis

Cont...

- Adenylate cyclase/hemolysin
 - a bifunctional toxin
 - activated in the target mammalian cell by intracellular calmodulin
 - catalyzes the conversion of endogenous adenosine triphosphate (ATP) to cAMP in eukaryotic cells
 - inhibits leukocyte chemotaxis, phagocytosis, and killing
 - initial protection of the bacteria during the early stages of disease

Cont...

- Dermonecrotic toxin

- at low doses causes vasoconstriction of peripheral blood vessels in mice
→ localized ischemia, the movement of leukocytes to extravascular spaces, and haemorrhage
- At high doses causes fatal reactions in mice
- localized tissue destruction in human infections

- Tracheal cytotoxin

- At low concentrations, it causes ciliostasis (inhibition of cilia movement)
- at the higher concentrations produced later in the infection, it causes extrusion of ciliated cells

Cont...

- interferes with deoxyribonucleic acid (DNA) synthesis
 - impairing the regeneration of damaged cells
- disrupts the normal clearance mechanisms in the respiratory tree
 - the characteristic cough associated with pertussis
- the release of the cytokine interleukin-1 (IL-1)
 - fever
- two distinct lipopolysaccharides, one with lipid A and the other with lipid X
 - activate the alternate complement pathway and stimulate cytokine release
 - Their role in the disease process is unknown

Mycoplasma pneumoniae

- the **smallest free-living bacteria**
- unique among bacteria because they **do not have a cell wall**, and their cell membrane contains **sterols**
- other cell-wall-deficient bacteria (called L forms) do not have sterols in their cell membrane and can form cell walls under the appropriate growth conditions
- resistant to penicillins, cephalosporins, vancomycin, and other antibiotics that interfere with synthesis of the cell wall
- form pleomorphic shapes varying from 0.2 to 0.3 μm coccoid forms to rods 0.1 to 0.2 μm in width and 1 to 2 μm long

Cont...

- pass through the 0.45- μm filters used to remove bacteria from solutions, which was why the mycoplasmas were originally thought to be viruses
- divide by binary fission (typical of all bacteria), grow on artificial cell-free media, and contain both ribonucleic acid (RNA) and deoxyribonucleic acid (DNA)
- a **strict aerobe** and require exogenous sterols supplied by animal serum added to the growth medium
- grow slowly and most form small colonies that are difficult to detect without extended incubation
- the major antigenic determinants are membrane glycolipids and proteins
 - cross-react with human tissues and other bacteria

Cont...

Pathogenesis

- ✓ an extracellular pathogen that adheres to the respiratory epithelium by means of a specialized attachment structure that forms at one end of the cell
 - consists of a complex of adhesions proteins, with the **P1 adhesin** the most important
 - interact specifically with sialated glycoprotein receptors at the base of cilia on the epithelial cell surface (and on the surface of erythrocytes)
- ✓ Ciliostasis then occurs, after which first the cilia, then the ciliated epithelial cells, are destroyed
 - interferes with the normal clearance of the upper airways and permits the lower respiratory tract to become contaminated with microbes and mechanically irritated

Cont...

- the persistent cough present in patients with symptomatic disease
- ✓ functions as a superantigen, stimulating inflammatory cells to migrate to the site of infection and release cytokines, initially tumor necrosis factor- α (TNF- α) and interleukin-1 (IL-1) and later, IL-6
 - both the clearance of the bacteria and the observed disease

Chlamydophila pneumoniae

- first isolated from the conjunctiva of a child in Taiwan
- initially considered a psittacosis strain, because the morphology of the inclusions produced in cell culture was similar
- subsequently shown that the Taiwan isolate (TW-183) was related serologically to a pharyngeal isolate, designated AR-39, and was unrelated to psittacosis strains
- This new organism was initially called TWAR, then classified as *Chlamydia pneumoniae*, and finally placed in the new genus *Chlamydophila*
- no animal reservoir

Cont...

- elementary bodies are resistant to many harsh environmental factors
- lack the rigid peptidoglycan layer
- their central dense core is surrounded by a cytoplasmic membrane and a double-layer outer membrane
- The cell wall contains a lipopolysaccharide (LPS) that is common to all members of the family
 - weak endotoxin activity
- The **major outer membrane protein (MOMP)** in the cell wall is an important structural component of the outer membrane and is unique for each species

Cont...

- the *C. pneumoniae* MOMP is homogeneous, and only a single serovar has been described
- A second, highly conserved outer membrane protein, **OMP 2**, is shared by all members of the family Chlamydiaceae
 - extensive disulfide cross-links that provide the stability in the elementary bodies
- The elementary bodies cannot replicate but are infectious
 - bind to receptors on host cells and stimulate uptake by the infected cell

Cont...

- In this intracellular location, the elementary bodies convert into reticulate bodies, the metabolically active, replicating chlamydial form
- the extensive cross-linked proteins are absent in reticulate bodies
- osmotically fragile
- protected by their intracellular location
- replicate by means of a unique growth cycle that occurs within susceptible host cells
- Within 6 to 8 hours after entering the cell, the elementary bodies reorganize into the larger (800 to 1000 nm), metabolically active reticulate bodies
- **energy parasites**
 - use host cell adenosine triphosphate for their energy requirements

Cont...

- Some strains may also depend on the host to provide specific amino acids
- The reticulate bodies replicate by binary fission
- histologic stains can readily detect the phagosome with accumulated reticulate bodies, called an **inclusion**
- Approximately 18 to 24 hours after infection, the reticulate bodies begin reorganizing into the smaller elementary bodies
- between 48 and 72 hours, the cell ruptures and then releases the infectious bacteria

Cont...

- a **human pathogen** that causes sinusitis, pharyngitis, bronchitis, and pneumonia
- transmitted person to person by respiratory secretions
- asymptomatic or mild, causing a persistent cough and malaise
- do not require hospitalization
- More severe respiratory tract infections typically involve a single lobe of the lungs
 - cannot be differentiated from other atypical pneumonias, such as those caused by *Mycoplasma pneumoniae*, *Legionella pneumophila*, and respiratory viruses

Cont...

GENERAL DIAGNOSTIC APPROACHES

- ❖ When a viral etiology is sought, the usual method of obtaining a specific diagnosis is by inoculation of cell cultures with material from the nasopharynx and throat
- ❖ In bacterial infections, the approaches noted below are valuable

Epiglottitis

- *H. influenzae type b* produces an associated bacteremia in 85% of cases or more

Cont...

- Attempts to obtain cultures from the epiglottis or throat may provoke acute reflex airway obstruction in patients who have not undergone intubation to ensure proper ventilation
- the yield is lower than that of blood culture
- other bacterial agents that cause epiglottitis can often be isolated from the blood
- *Corynebacterium diphtheriae* infection, in which cultures of the nasopharynx or pharynx are required

Cont...

Laryngotracheitis and Laryngotracheobronchitis

- Although most cases of laryngotracheitis and laryngotracheobronchitis have a viral etiology, a severe purulent process is seen occasionally
 - as acute **bacterial tracheitis**
- Gram staining and culture of sputum, or better yet, of purulent secretions obtained by direct laryngoscopy, help establish the causative agent
- Blood cultures are again useful in such cases when a bacterial etiology is suspected

Cont...

Acute Bronchitis

- A major bacteriologic consideration in acute bronchitis, especially in infants and preschool children, is *Bordetella pertussis*
- Deep nasopharyngeal cultures plated on the appropriate media constitute the best specimens
- Gram staining and examination of nasopharyngeal smears by direct fluorescent antibody methods are also useful adjuncts to establishing the diagnosis
- When purulent sputum is produced, Gram staining and culture may be useful in suggesting other bacterial causes

Cont...

- *M. pneumoniae* and *C. pneumoniae* infections, which are usually diagnosed by serologic testing of acute and convalescent sera

GENERAL PRINCIPLES OF MANAGEMENT

- ✓ The primary initial concern is ensuring an adequate airway
- ✓ In proved or suspected bacterial infections, specific antimicrobial therapy is required
- ✓ antitoxin administration in diphtheria, may also be necessary

Cont...

LOWER RESPIRATORY TRACT INFECTION

- ❖ develops with invasion and disease of the lung, including the alveolar spaces and their supporting structure, the interstitium, and the terminal bronchioles
- ❖ occur
 - extension of a middle respiratory tract infection
 - aspiration of pathogens past the upper airway defenses
 - less commonly by hematogenous spread from a distant site such as an abscess or an infected heart valve

Cont...

CLINICAL FEATURES

Acute Pneumonia

- an infection of the lung parenchyma that develops over hours to days and, if untreated, runs a natural course lasting days to weeks
- The onset may be gradual, with malaise and slowly increasing fever, or sudden, as with the bed-shaking chill associated with the onset of pneumococcal pneumonia
- The only early symptom referable to the lung may be cough
 - bronchial irritation

Cont...

- In adults the cough becomes productive of **sputum**
- the sputum may be blood streaked, rusty in color, or foul smelling
- Labored or difficult breathing (dyspnea) and rapid respiratory rate are signs of increasing loss of alveolar air-exchange surface through spread of exudate
- Chest pain from inflammatory involvement of the pleura is common

Cont...

Chronic Pneumonia

- ❑ has a slow insidious onset that develops over weeks to months and may last for weeks or even years
- ❑ initial symptoms
 - fever, chills, and malaise
 - develop more slowly
- ❑ Cough can develop early or late in the illness
- ❑ appetite and weight loss, insomnia and night sweats are common
- ❑ Cough and sputum production may be the first indication of a vague constitutional illness referable to the lung
- ❑ Bloody sputum (hemoptysis), dyspnea and chest pain

Cont...

❑ **Empyema** is a purulent infection of the pleural space

Lung Abscess

- ✓ a complication of acute or chronic pneumonia
- ✓ occur as part of a chronic process or as an extension of an acute, destructive pneumonia
- ✓ not specific, resemble those of chronic pneumonia or an acute pneumonia that has failed to resolve
- ✓ Persistent fever, cough, and the production of foul-smelling sputum are typical
- ✓ Multiple abscesses
 - blood-borne infection

Cont...

COMMON ETIOLOGIC AGENTS

- The etiology of acute pneumonia is strongly dependent on age
- More than 80% of pneumonias in infants and children
 - viruses
- less than 10 to 20% of pneumonias in adults
 - viral
- Influenza and other viruses, however, may provide the initial predisposition toward bacterial infection
- At any age, the pneumococcus is the most common bacterial cause of acute pneumonia

Cont...

- Pneumonia caused by bacteria such as enteric Gram-negative rods, *Pseudomonas*, and *Legionella* is primarily limited to patients with serious debilitating underlying disease or as a complication of hospitalization and its procedures (nosocomial infection)
- Gram-negative infections other than *Haemophilus* are rare in children unless they have cystic fibrosis or immunodeficiency
- *C. trachomatis* is almost exclusively limited to infants less than 7 months of age
- *C. pneumoniae* commonly affects school children and young adults, producing both bronchitis and pneumonia

Cont...

- Lung abscess and empyema follow infections with the more destructive organisms or aspiration of mixed anaerobic flora from the oropharynx

Major Causes of Lower Respiratory Tract Infection

SYNDROME	VIRUSES	COMMON BACTERIA	FUNGI	OTHER AGENTS
Acute pneumonia	Influenza, parainfluenza, adenovirus, respiratory syncytial virus (infants)	<i>S. pneumoniae</i> , <i>S. aureus</i> , <i>H. influenzae</i> , Enterobacteriaceae, <i>Legionella</i> , mixed anaerobes (aspiration), <i>P. aeruginosa</i>	<i>Candida albicans</i> <i>Aspergillus species</i>	<i>Mycoplasma pneumoniae</i> , <i>Pneumocystis carinii</i> , <i>Chlamydia trachomatis</i> (infants), <i>C. pneumoniae</i>
Chronic pneumonia	Rare	<i>Mycobacterium tuberculosis</i> , other mycobacteria, <i>Nocardia</i>	<i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i> , <i>Histoplasma capsulatum</i> , <i>Cryptococcus neoformans</i>	<i>Paragonimus westermani</i>

SYNDROME	VIRUSES	COMMON BACTERIA	FUNGI	OTHER AGENTS
Lung abscess	None	Mixed anaerobes, <i>Actinomyces</i> , <i>Nocardia</i> , <i>S. aureus</i> , Enterobacteriaceae, <i>P. aeruginosa</i>	<i>Aspergillus</i> <i>species</i>	<i>Entamoeba</i> <i>histolytica</i>
Empyema	None	Mixed anaerobes, <i>S. aureus</i> , <i>S. pneumoniae</i> , Enterobacteriaceae, <i>P. aeruginosa</i>	Rare	

Enterobacteriaceae

Physiology and Structure

- moderately sized gram-negative rods
- share a common antigen (enterobacterial common antigen), are either motile with peritrichous flagella or nonmotile and do not form spores
- can grow rapidly, aerobically and anaerobically, on a variety of nonselective (e.g., blood agar) and selective (e.g., MacConkey agar) media
- have simple nutritional requirements, ferment glucose, reduce nitrate, and are catalase positive and oxidase negative
- The absence of cytochrome oxidase activity is used to distinguish the Enterobacteriaceae from many other fermentative and nonfermentative gram-negative rods

Cont...

- Characteristics of the organisms' colonies on different media
 - identify common members of the family Enterobacteriaceae
- the ability to ferment lactose
 - differentiate lactose-fermenting strains (e.g., *Escherichia*, *Klebsiella*, *Enterobacter*, *Citrobacter*, and *Serratia* spp.; pink-purple colonies on MacConkey agar) from strains that do not ferment lactose or do so slowly (e.g., *Proteus*, *Salmonella*, *Shigella*, and *Yersinia* spp.; colorless colonies on MacConkey agar)

Cont...

- have prominent capsules (e.g., most *Klebsiella*, some *Enterobacter* and *Escherichia* strains)
- a loose fitting, diffusible slime layer surrounds other strains
- The heat-stable lipopolysaccharide (LPS) is the major cell wall antigen
 - three components: the outermost somatic O polysaccharide, a core polysaccharide common to all Enterobacteriaceae, and lipid A
- The core polysaccharide
 - classifying an organism as a member of the Enterobacteriaceae
- the O polysaccharide
 - the epidemiologic classification of strains within a species

Cont...

- the lipid A
 - endotoxin activity, an important virulence factor
- The epidemiologic (serologic) classification
 - three major groups of antigens: somatic O polysaccharides, capsular K antigens (type-specific polysaccharides), and the flagellar H proteins
- Specific O antigens are present in each genus, although cross-reactions between closely related genera are common (e.g., Salmonella with Citrobacter, Escherichia with Shigella)
- The antigens are detected by agglutination with specific antibodies
- The heat-labile K antigens are not commonly used for strain typing but are important because they may interfere with detection of the O antigens

Cont...

- Boiling of the organism to remove the heat-labile K antigen and expose the heat-stable O antigen circumvents this problem
- The H antigens are heat-labile flagellin proteins
 - absent from a cell, or they may undergo antigenic variation and be present in two phases
- are motile, with the exception of the common isolates *Klebsiella*, *Shigella*, and *Yersinia*
- The motile strains possess peritrichous flagella
- possess fimbriae (also referred to as pili)
 - subdivided into two general classes: chromosomally mediated common fimbriae and sex pili that are encoded on conjugative plasmids

Cont...

- The common fimbriae
 - the ability of bacteria to adhere to specific host cell receptors
- the sex or conjugative pili
 - genetic transfer between bacteria

Pathogenesis

- ❑ Numerous virulence factors
- ❑ Some are common to all genera, and others are unique to specific virulent strains

Endotoxin

- ❖ shared among aerobic and some anaerobic gram-negative bacteria

Cont...

- ❖ depends on the lipid A component of LPS, which is released at cell lysis
- ❖ Many of the systemic manifestations of gram-negative bacterial infections are initiated by endotoxin activation of complement, release of cytokines, leukocytosis, thrombocytopenia, disseminated intravascular coagulation, fever, decreased peripheral circulation, shock, and death

Capsule

- protected from phagocytosis by the hydrophilic capsular antigens, which repel the hydrophobic phagocytic cell surface
 - interfere with the binding of antibodies to the bacteria and are poor immunogens or activators of complement
- The protective role of the capsule is diminished, however, if the patient develops specific anticapsular antibodies

Cont...

- The expression of capsular K and flagellar H antigens is under the genetic control of the organism
 - can be alternately expressed or not expressed (phase variation), a feature that protects the bacteria from antibody-mediated cell death

Type III Secretion Systems

- ✓ A variety of bacteria (e.g., Yersinia, Salmonella, Shigella, enteropathogenic Escherichia, Pseudomonas, and Chlamydia) have a common effector system for delivering their virulence factors into targeted eukaryotic cells
- ✓ approximately 20 proteins that facilitate secretion of bacterial virulence factors when the bacteria come into contact with the host cells

Cont...

- ✓ the virulence factors and their effects differ among the various gram-negative rods
- ✓ the general mechanism by which the virulence factors are introduced is the same

Sequestration of Growth Factors

- Nutrients are provided to the organisms in enriched culture media, but the bacteria must become nutritional scavengers when growing in vivo
- Iron is an important growth factor required by bacteria, but it is bound in heme proteins (e.g., hemoglobin, myoglobin) or in iron-chelating proteins (e.g., transferrin, lactoferrin)

Cont...

- The bacteria counteract the binding by producing their own competitive siderophores or iron-chelating compounds (e.g., enterobactin, aerobactin)
- Iron can also be released from host cells by hemolysins produced by the bacteria

Resistance to Serum Killing

- many bacteria can be rapidly cleared from blood
- virulent organisms capable of producing systemic infections are often resistant to serum killing
- the bacterial capsule can protect the organism from serum killing
- other factors prevent the binding of complement components to the bacteria and subsequent complement-mediated clearance

Cont...

Antimicrobial Resistance

- As rapidly as new antibiotics are introduced, organisms can develop resistance to them
 - encoded on transferable plasmids and exchanged among species, genera, and even families of bacteria

Pseudomonas aeruginosa

Physiology and Structure

- ❖ Straight or slightly curved gram-negative bacilli
- ❖ Polar flagella that render them motile
- ❖ Nonfermentative, utilizing carbohydrates through respiratory metabolism with oxygen the terminal electron acceptor
- ❖ Obligate aerobes
- ❖ Grow anaerobically using nitrate or arginine as an alternate electron acceptor
- ❖ Cytochrome oxidase
 - differentiate them from the Enterobacteriaceae

Cont...

- ❖ Some strains appear mucoid because of the abundance of a polysaccharide capsule
 - particularly common in patients with cystic fibrosis
- ❖ Some species produce diffusible pigments (e.g., pyocyanin [blue], pyoverdine [yellow-green], and pyorubin [reddish-brown]) that give them a characteristic appearance in culture

Pathogenesis

- many virulence factors
 - structural components, toxins, and enzymes
- the delivery system used by *Pseudomonas*, the type III secretion system, is particularly effective in injecting toxins into the host cell

Cont...

- Despite the diversity of virulence factors, most experts believe that multiple factors must work together for *P. aeruginosa* to cause disease

Adhesins

- ✓ Adherence to host cells is critical for establishing infection
- ✓ At least four structural components on the surface of *P. aeruginosa* facilitate this adherence: (1) flagella, (2) pili, (3) lipopolysaccharide (LPS), and (4) alginate
- ✓ Flagella and pili
 - motility in *P. aeruginosa*
- ✓ lipid-A
 - endotoxin activity

Cont...

- ✓ Alginate is a mucoid exopolysaccharide that forms a prominent capsule on the bacterial surface and protects the organism from phagocytosis and the activity of antibiotics

Secreted Toxins and Enzymes

Exotoxin A (ETA)

- disrupts protein synthesis by blocking peptide chain elongation in eukaryotic cells, much like the diphtheria toxin produced by *Corynebacterium diphtheria*
 - structurally and immunologically different, and exotoxin A is less potent than diphtheria toxin
- Dermatonecrosis that occurs in burn wounds, corneal damage in ocular infections, and tissue damage in chronic pulmonary infections

Cont...

- immunosuppressive

Pyocyanin

- A blue pigment
- Production of superoxide and hydrogen peroxide
- stimulates interleukin-8 (IL-8) release, leading to enhanced attraction of neutrophils

Pyoverdin

- yellow-green pigment
- a siderophore that binds iron for use in metabolism
- regulates secretion of other virulence factors, including exotoxin A

Cont...

Elastases

- ❑ LasA (serine protease) and LasB (zinc metalloprotease) act synergistically to degrade elastin
 - damage to elastin-containing tissues
 - the lung parenchymal damage and hemorrhagic lesions (**ecthyma gangrenosum**)
- ❑ Degrade complement components and inhibit neutrophil chemotaxis and function
 - further spread and tissue damage in acute infections

Cont...

Alkaline protease

- ❖ Contributes to tissue destruction and spread of *P. aeruginosa*
- ❖ Interferes with the host immune response

Phospholipase C

- Breaks down lipids and lecithin, facilitating tissue destruction

Exoenzymes S and T

- ✓ Extracellular toxins
- ✓ Introduced into their target eukaryotic cells by the type III secretion system
 - epithelial cell damage occurs, facilitating bacterial spread, tissue invasion, and necrosis

Cont...

Antibiotic resistance

- Inherently resistant to many antibiotics
- Mutate to even more resistant strains during therapy
- the mutation of porin proteins constitutes the major mechanism of resistance
- Penetration of antibiotics into the pseudomonad cell is primarily through pores in the outer membrane
- If the proteins forming the walls of these pores are altered to restrict flow into the cell, resistance to many classes of antibiotics can develop simultaneously
- produces a number of different β -lactamases that can inactivate many β -lactam antibiotics (e.g., penicillins, cephalosporins, and carbapenems)

Mycobacterium tuberculosis

- Mycobacterium
- Nonmotile, non-spore-forming, aerobic bacilli
- The cell wall is rich in lipids
 - surface hydrophobic
 - resistant to common laboratory stains
- cannot be decolorized with acid solutions
 - **acid-fast bacilli**
- Grow slowly

Cont...

Pathogenesis

- Penetrate into unactivated alveolar macrophages
- Inhibit acidification of the phagosome and subsequent phagosome-lysosome fusion and proceed to replicate freely
- The infected phagocytic cells are eventually destroyed
 - further cycles of phagocytosis by macrophages, mycobacterial replication, and cell lysis
- Circulating macrophages and lymphocytes are attracted to the infectious focus by the bacilli, cellular debris, and host chemotactic factors (C5a)

Cont...

- Infected macrophages

- local lymph nodes as well as into the blood stream and other tissues (e.g., bone marrow, spleen, kidneys, CNS)

- Histologic signs

- components of the host response to the infection rather than specific virulence factors

Cont...

GENERAL DIAGNOSTIC APPROACHES

- degree of difficulty
 - the number of organisms produced in respiratory secretions
 - whether the causative species is normally found in the oropharyngeal flora
 - how easily it is grown
- In the presence of typical clinical findings, the isolation of influenza virus from the throat or of *M. tuberculosis* from sputum is sufficient for diagnosis of influenza or tuberculosis
- The same cannot be said for *S. pneumoniae* and most bacterial pathogens

Cont...

- The examination of expectorated sputum has been the primary means of diagnosing the causes of bacterial pneumonia
 - advantages and disadvantages
- Advantages
 - ease of collection and absence of risk to the patient
- Disadvantage
 - confusion that results from contamination of the sputum with oropharyngeal flora in the process of expectoration and excessive contamination with saliva

Cont...

- The quality of a sputum sample can be enhanced by collection early in the morning and careful instruction of the patient
- Polymorphonuclear leukocytes and large numbers of a single morphologic type of organism are typical findings in sputum from patients with bacterial pneumonia
- Squamous epithelial cells from the oropharynx and a mixed bacterial population are characteristic of saliva
- most specimens are a mixture of both, which makes interpretation more difficult
- more than 10 to 25 squamous epithelial cells per low-power microscopic field are evidence of excessive salivary contamination, and such specimens should not be cultured

Cont...

- the direct Gram smear is crucial to the use of expectorated sputum for diagnosis of acute bacterial pneumonia
- The smear may be useful in the absence of cultural results, but cultures are useless without a Gram smear to assess specimen quality
- a more direct collection from the lung using methods that bypass the oropharyngeal flora
 - in patients who are not producing sputum or in cases where analysis of expectorated sputum has been inconclusive
 - transtracheal aspiration, bronchoalveolar lavage (BAL), direct aspiration, and open biopsy

Cont...

- transtracheal aspiration
 - in diagnosis of both pneumonia and lung abscess
- BAL
 - demonstration of *Pneumocystis carinii* that were previously only seen in open lung biopsies
- Bacteremia may occur in acute pneumonia, particularly in its early stages
- A blood culture should be part of the evaluation of every acute pneumonia

Cont...

LABORATORY EXAMINATION OF SPUTUM

Day 1

Specimens for immediate attention: When pneumonia or bronchopneumonia is suspected, the sputum must be cultured as soon as possible because organisms such as *H. influenzae* and *S. pneumoniae* do not survive well in specimens

1 Describe the appearance of the specimen

- Purulent: Green-looking, mostly pus
- Mucopurulent: Green-looking with pus and mucus
- Muroid: Mostly mucus
- Mucosalivary: Mucus with a small amount of saliva

Cont...

- When the sputum contains blood, this must also be reported

2 Examine the specimen microscopically

Gram smear

Examine the smear for pus cells and predominant bacteria

- ☐ Gram positive diplococci (capsulated) → *S. pneumoniae*
- ☐ Gram positive cocci in groups → *S. aureus*
- ☐ Gram negative rods and cocco-bacilli → *H. influenza*
- ☐ Gram negative capsulated rods → *K. pneumoniae*
- ☐ Gram negative diplococci in and between pus cells
→ *M. catarrhalis*

Cont...

❑ *Note:* When pus cells are present but no bacteria are seen in a Gram stained smear, this may indicate the presence of microorganisms such as *M. tuberculosis*, *Chlamydomphila pneumoniae*, *Mycoplasma pneumoniae*, *Legionella pneumophila* or viruses

Ziehl-Neelsen smear to detect AFB

- ❖ the chances of detecting AFB in sputum smears are significantly increased when sputum is first treated with 5% v\ v sodium hypochlorite (NaOC1), i.e. bleach, followed by centrifugation or overnight sedimentation
- ❖ *Note:* Up to three specimens may need to be examined to detect *M. tuberculosis* in sputum.

Cont...

3 Culture the specimen

Blood agar and chocolate agar

- Inoculate the washed sputum on plates of:
 - Blood agar
 - Chocolate (heated blood) agar
- Add an optochin disc to the blood agar plate within the area of 2nd spread
 - *S. pneumoniae*

ADDITIONAL

Culture of sputum for *M. tuberculosis*

Cont...

Day 2 and Onwards

4 Examine and report the cultures

Blood agar and chocolate agar cultures

- ✓ Streptococcus pneumoniae sensitive to optochin
- ✓ Haemophilus influenzae
- ✓ Staphylococcus aureus

Less frequently isolated pathogens

- *K. pneumoniae*, *P. aeruginosa*, *M. catarrhalis*, *S. pyogenes*, *Proteus species*, *Candida albicans*

Cont...

Antimicrobial susceptibility testing

- performed only when the amount of cultural growth of a pathogen is significant
- Strains of *S. pneumoniae* should be tested on blood agar for susceptibility to penicillin, tetracycline, and erythromycin
- *H. influenzae* strains should be tested for betalactamase production and susceptibility to ampicillin, tetracycline, and co-trimoxazole

Cont...

GENERAL PRINCIPLES OF MANAGEMENT

- similar to those of middle tract infections
- Drainage or surgical measures are needed more often as adjuncts to antimicrobial therapy in cases of chronic pneumonia, lung abscess, and empyema
- When bacterial infection is considered, empirical therapy is usually given until the results of cultures and antimicrobial susceptibility tests are available
- Treatment may vary from penicillin alone for *S. pneumoniae*, to multiple drugs for a debilitated or immunocompromised patient, in whom the possibilities are much broader

Cont...

- Mycobacterial infections

- Complex and controversial
- Patients must take multiple antibiotics for an extended period (e.g., a minimum of 6 to 9 months)
- A traditional regimen
 - combination of isoniazid and rifampin for 9 months
- Pyrazinamide and either ethambutol or streptomycin should be given with this regimen until susceptibility results are available